

How GPT-6 Astra Works: Self-Attention and LLMs

Self-Attention

1. Sentence – A sentence is a group of words.
2. Tokenization – The text is divided or broken down into smaller individual parts called tokens. These can be words or smaller chunks of words.
3. Word Embedding – Word embeddings represent how much words are related to each other. Words are converted into vectors, and each word is assigned a numerical representation. These vectors can have n dimensions (or any specified number of dimensions).
Example: GPT-6 Astra has 16,384 dimensions.
4. Comparing Words – One word is compared with all the other words in the sentence. For this, we use the dot product.
5. Calculating Attention Scores – The dot products are used to calculate attention scores, which show how relevant one word is to the others.
6. Normalizing the Values – The attention scores are normalized using the softmax function, which converts them into values between 0 and 1. Once normalized, these values are called attention weights, representing the importance of each word.
7. Context Vector – The context vector is calculated by multiplying the attention weights by the input vectors and adding the results. This creates a representation that contains the important contextual information.

How LLMs Actually Work

An LLM does one main thing: Read → Predict → Append → Repeat.

1. Tokens

Text is converted into tokens. Tokens are small chunks of text, and sometimes a word is divided into multiple tokens.

- 1 token $\approx$ 4 characters
- 100 tokens $\approx$ 75 words

The output generation process can take more time than processing the input because the input is processed all at once (in parallel), while the output is generated one token at a time (sequentially).

Example: GPT-6 Astra has 200,000 tokens in its vocabulary.

Pricing								
Flagship models								
Our latest models								
Prices per 1M tokens.								
Model	Short context				Long context			
	Input	Cached input	Cache writes	Output	Input	Cached input	Cache writes	Output
gpt-6-astra	\$10.00	\$1.00	\$12.50	\$50.00	\$20.00	\$2.00	\$25.00	\$75.00
gpt-5.6-sol	\$4.00	\$0.40	\$5.00	\$20.00	\$8.00	\$0.80	\$10.00	\$30.00
gpt-5.6-terra	\$2.00	\$0.20	\$2.50	\$12.00	\$4.00	\$0.40	\$5.00	\$18.00
gpt-5.6-luna	\$0.20	\$0.02	\$0.25	\$1.20	\$0.40	\$0.04	\$0.50	\$1.80

2. Prediction Loop

The model predicts the next word or token in a sentence. That predicted token is then added to the existing text, and the updated text is sent back to the LLM. The model then predicts the next token again.

This process continues repeatedly until the complete response is generated.

Read → Predict → Append → Repeat

3. Context Window

The context window is the maximum amount of information an LLM can process and keep available within a single interaction.

Example: GPT-6 Astra has a context window of 1,050,000 tokens (roughly 1.05 million tokens), meaning it can process approximately this much information at once.

Self-Attention

1 Sentence

A sentence is a group of words.

The cat sits on the mat.

2 Tokenization

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3 Word Embedding

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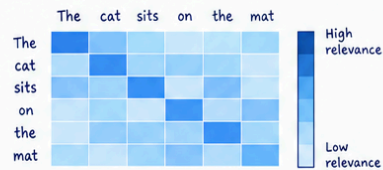
4 Comparing Words

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5 Calculating Attention Scores

The dot products are used to calculate attention scores, which show how relevant one word is to the others.



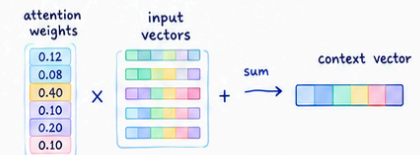
6 Normalizing the Values

The attention scores are normalized using the softmax function, which converts them into values between 0 and 1. Once normalized, these values are called attention weights, representing the importance of each word.



7 Context Vector

The context vector is calculated by multiplying the attention weights by the input vectors and adding the results. This creates a representation that contains the important contextual information.



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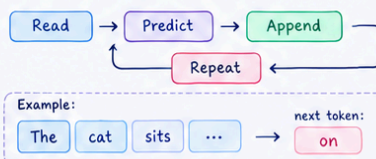
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